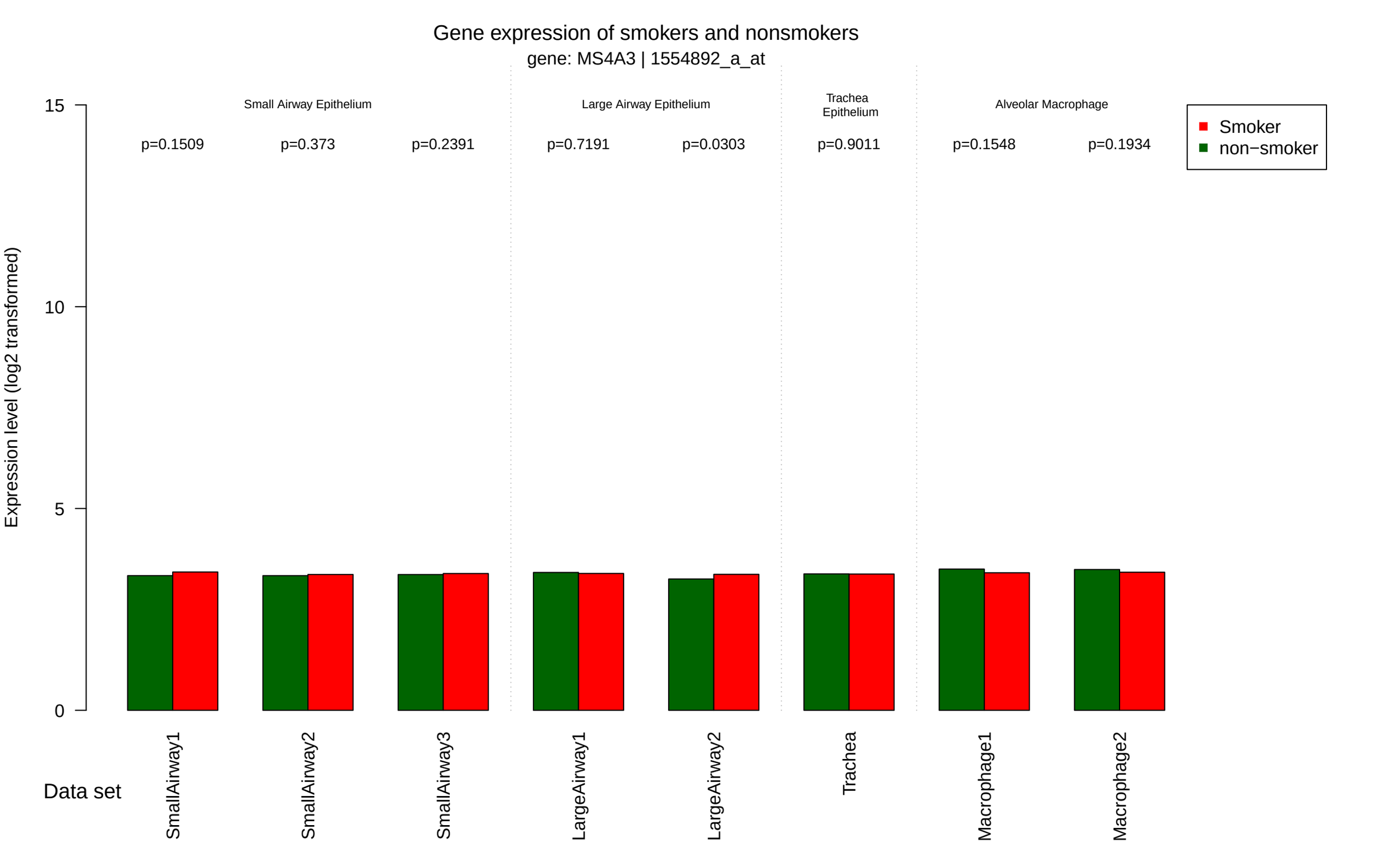
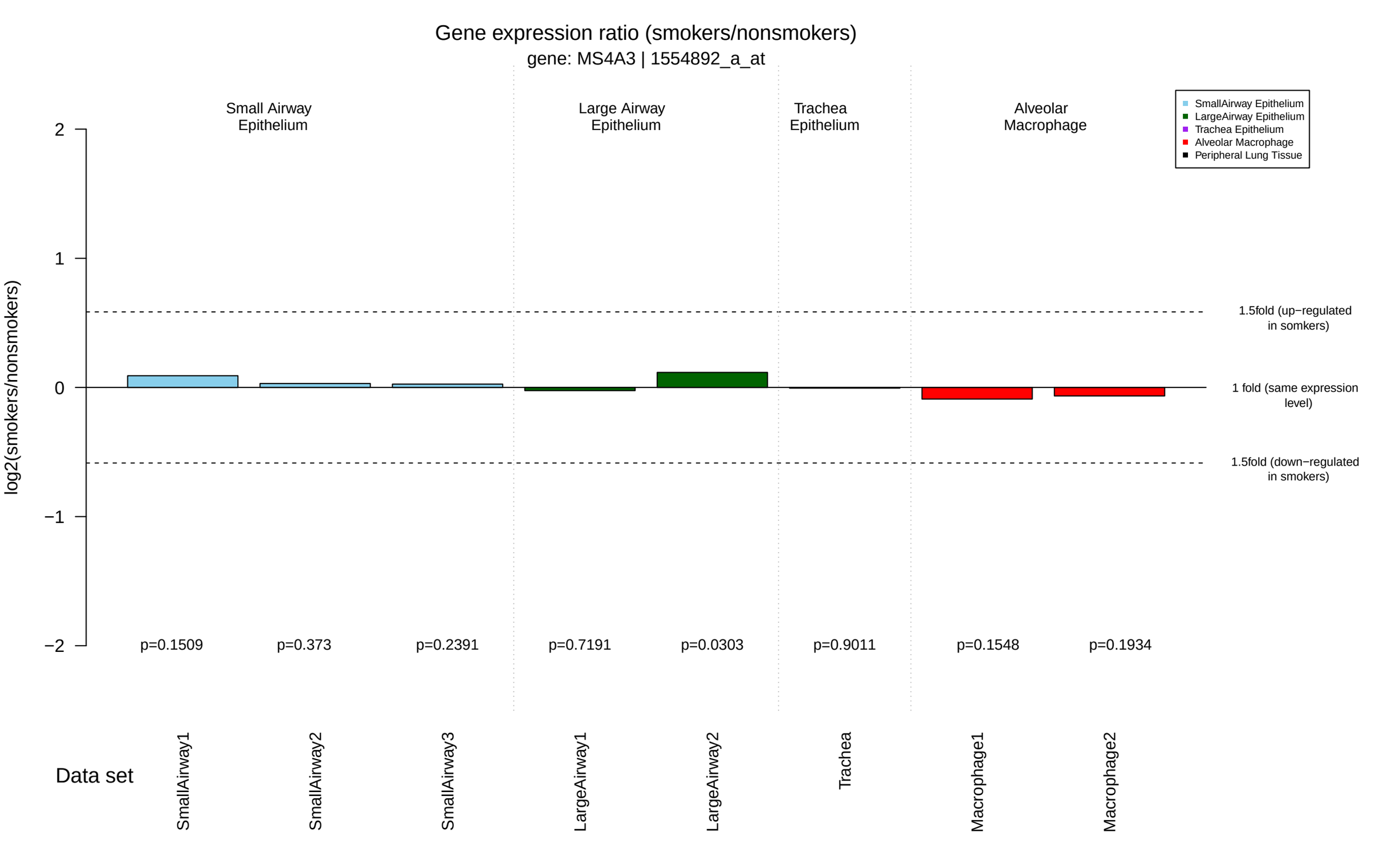
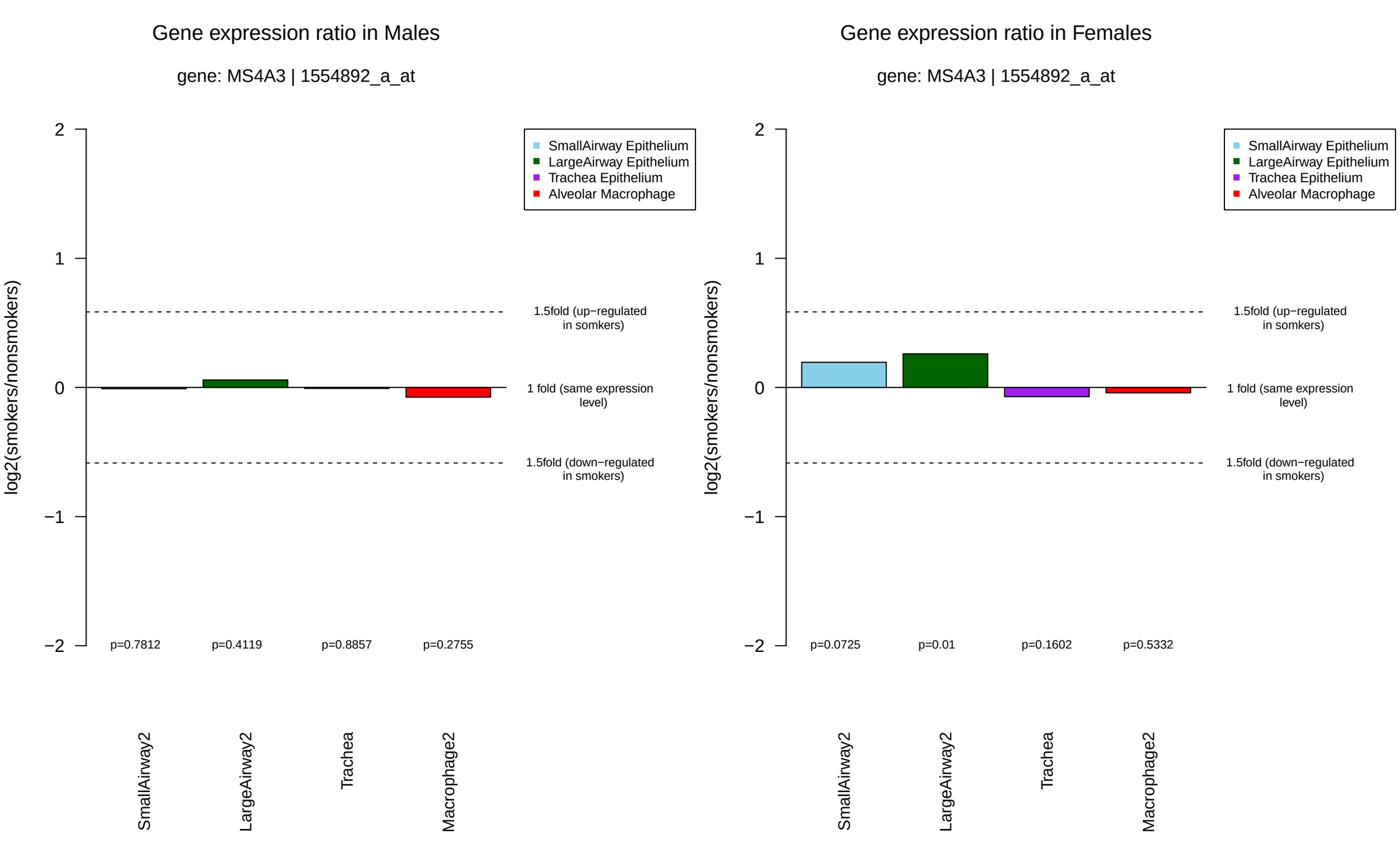
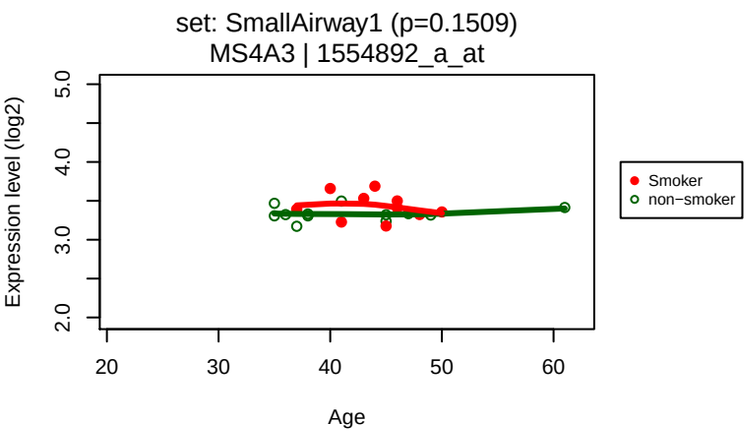
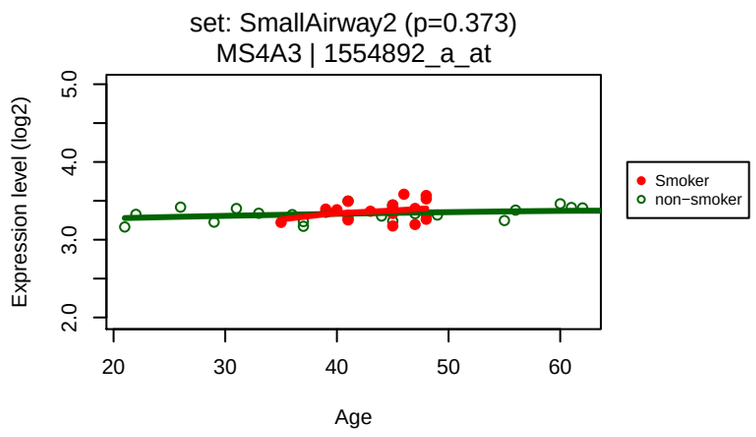
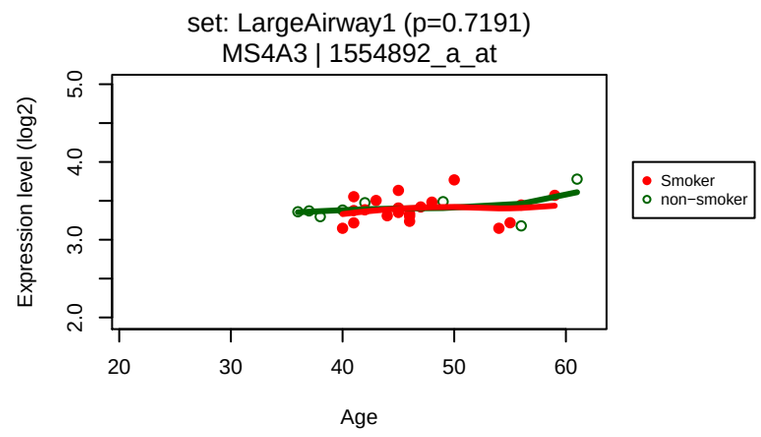

Gene expression of smokers and nonsmokers
gene: MS4A3 | 1554892_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1509
p=0.373
p=0.2391
p=0.7191
p=0.0303
p=0.9011
p=0.1548
p=0.1934
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

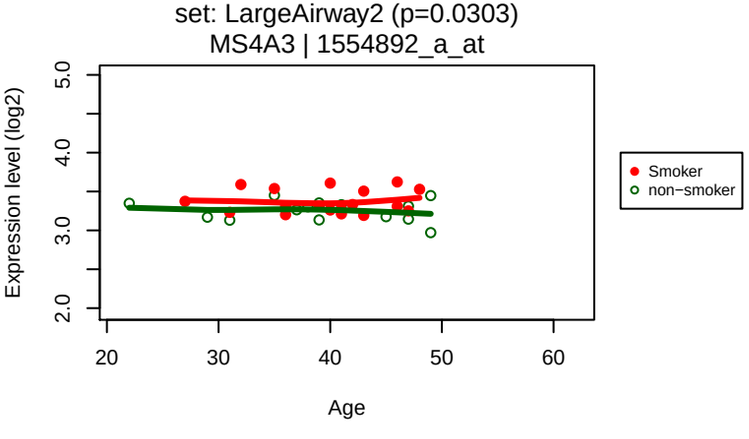

Gene expression ratio (smokers/nonsmokers)
gene: MS4A3 | 1554892_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1509
p=0.373
p=0.2391
p=0.7191
p=0.0303
p=0.9011
p=0.1548
p=0.1934
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

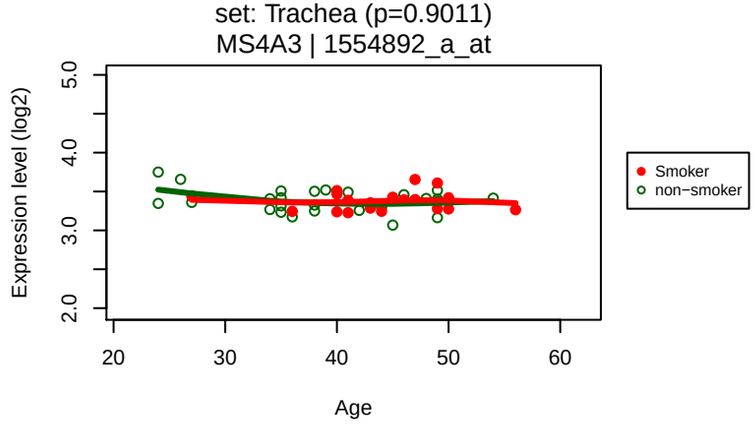

Gene expression ratio in Males
gene: MS4A3 | 1554892_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7812
p=0.4119
p=0.8857
p=0.2755
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MS4A3 | 1554892_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0725
p=0.01
p=0.1602
p=0.5332
SmallAirway2
LargeAirway2
Trachea
Macrophage2

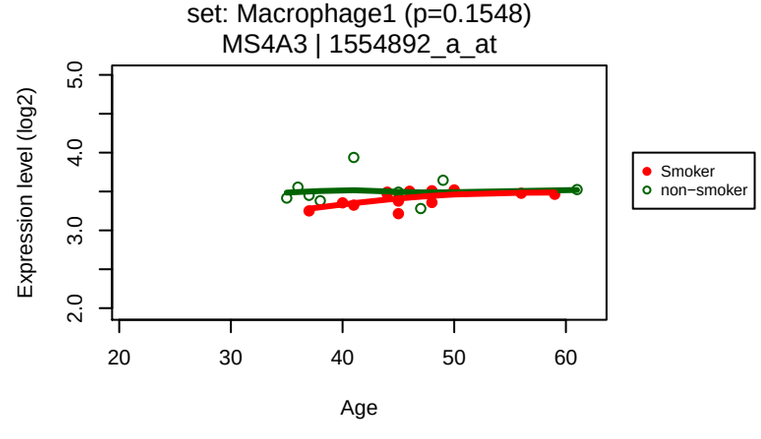

set: SmallAirway1 (p=0.1509)
MS4A3 | 1554892_a_at
Expression level (log2)
Age
Smoker
non−smoker

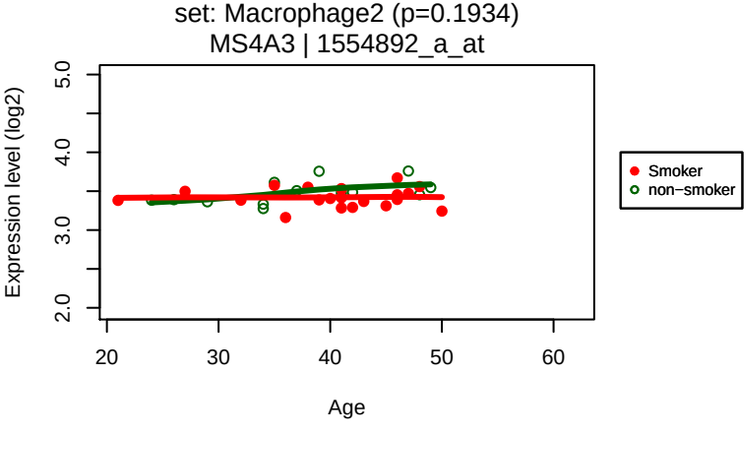

set: SmallAirway2 (p=0.373)
MS4A3 | 1554892_a_at
Expression level (log2)
Age
Smoker
non−smoker

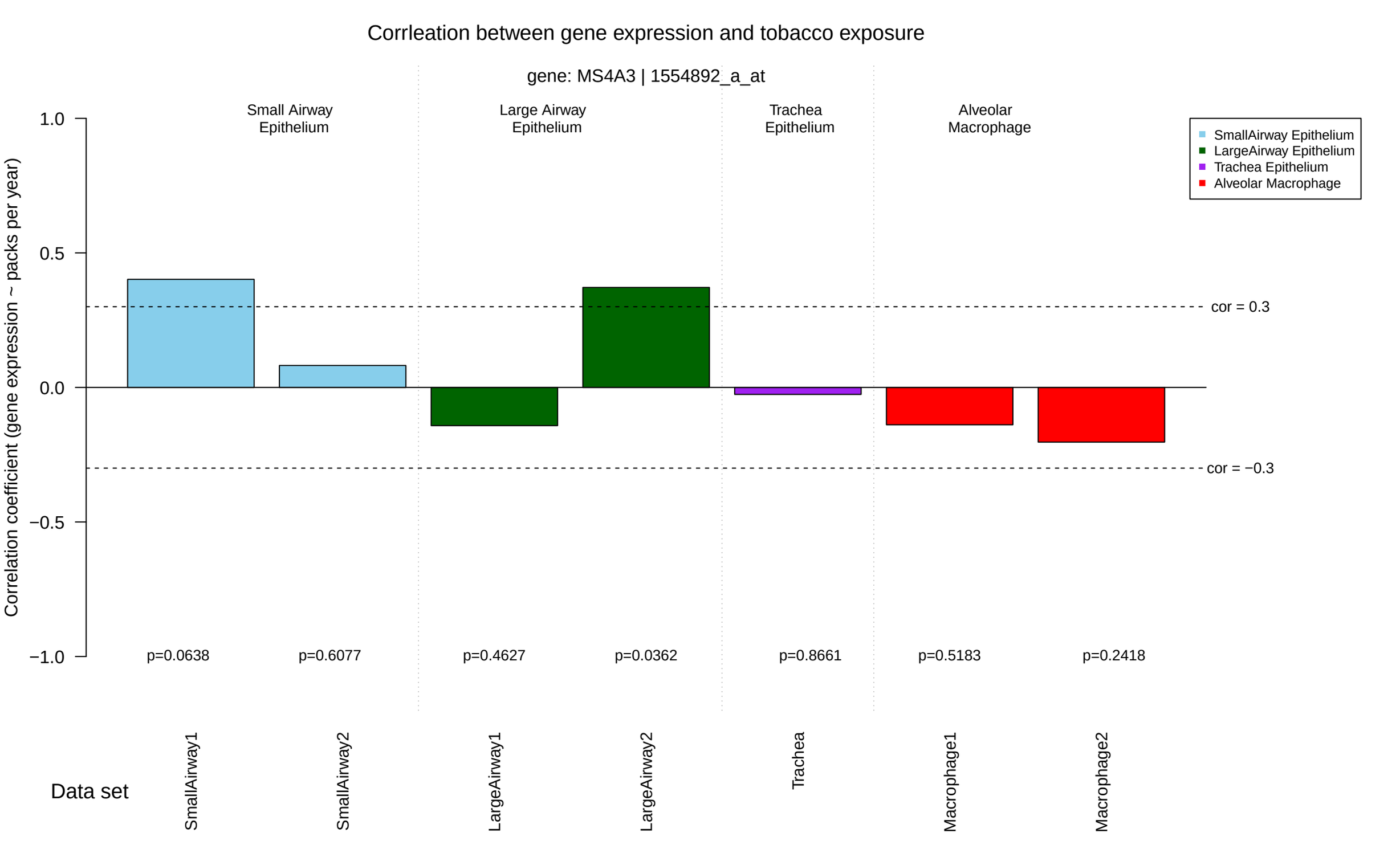

set: LargeAirway1 (p=0.7191)
MS4A3 | 1554892_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0303)
MS4A3 | 1554892_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9011)
MS4A3 | 1554892_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1548)
MS4A3 | 1554892_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1934)
MS4A3 | 1554892_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MS4A3 | 1554892_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0638
p=0.6077
p=0.4627
p=0.0362
p=0.8661
p=0.5183
p=0.2418
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2